

Michael Grady

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EDUCATION

University of Massachusetts Amherst

Spring 2026

Bachelor of Science in Mechanical Engineering

GPA: 3.99

- Dean's List (All Semesters), Minor in Engineering Management
- **Coursework:** Industrial Automation, Design of Mechanical Assemblies, Manufacturing Processes, Project Management

EXPERIENCE

Systems Intern – Surface Ship Program

May 2025 – May 2026

General Dynamics Mission Systems

Pittsfield, MA

- Supported the Ghost program prototype progression with mechanical design updates through use of SolidWorks and Creo
- Deployed config changes to proprietary Tactical Cross Domain Solutions to filter network traffic between secure domains
- Kept user docs up to date through technical writing; software instructions, user guides, test procedures, etc.

Manufacturing & Operations Intern

Summer 2024

UFP Technologies

Chicopee, MA

- Co-led 2 Kaizen workshops that drove ~\$250K in projected labor savings; managed action items that reduced cell footprint by 5% and cut operator/material travel by 50%
- Coordinated with MFG engineers and cell supervisors to design a one-piece flow and reduce WIP for a high-volume line
- Tracked material use and updated existing BOMs for transfer between Epicor & BOMCheck

Undergraduate Researcher

Spring 2025 – Fall 2025

Fluid-Structure Interactions Laboratory, UMass Amherst

- Completed research to investigate coupled-mode flutter in small-scale wind turbine blades
- Prototyped miniature blades based off the NREL 5MW turbine, cast and 3D printed to scale
- Used MATLAB to conduct frequency analysis of various airfoil models to determine oscillation frequencies

Mechanical Subteam

Fall 2025 – Spring 2026

UMass Robotics Research Team

- Worked with 10 students in the development of advanced robotic systems, including a quadrupedal robot
- Designed and analyzed the robot dog using OnShape, optimizing for efficiency and mechanical robustness
- Oversaw team meetings, coordinated schedules, and conducted actuation research to enhance system performance

Mechanical Team

2023 – 2025

UMass Mechatronics Team

- Designed mechanical assemblies for two competitions using CAD, including a precision size-sorting hopper and a golf robot
- Led 3D printing of prototypes and testing of final assemblies through full build cycles for both robots
- Placed 3rd out of 25+ teams (Ball-Bearing Sorter) and 8th out of 30+ teams (Miniature Golf Robot) at ASME competitions

Undergraduate Teaching Assistant

Spring 2024 – Spring 2026

Various Mechanical Engineering Courses, UMass Amherst

- Mentoring and guiding students through courses of varying difficulty, fostering a collaborative learning environment

PROJECTS

TTM Technologies Senior Capstone | *Systems Lead*

Fall 2025 – Spring 2026

- Engineered an Industry 4.0 computer vision system to replace an obsolete production asset for TTM Technologies
- Developed an HMI to automate image capture, processing, and printed circuitry registration calculations
- Integrated HMI with Ignition to collect registration error data and share across connected devices for real-time control
- Led cross-functional system integration by synchronizing electrical and mechanical workstreams

Industrial Recipe Control

Spring 2026

- Developed a real-time production tracking system by integrating Ignition Perspective with Studio 5000 via OPC UA
- Optimized manufacturing analytics by implementing cycle-time calculations and dynamic HMI status indicators
- Automatically logged cycle counts and process data into a SQL database using Ignition's built-in SQL Bridge module

PLC Control System

Fall 2025

- Designed and programmed an automated solution for an industrial mixing and coating process
- Programmed PLC logic for sequential control, developing the control narrative, safety interlocks, and alarm conditions
- Developed detailed Process Flow and Piping & Instrumentation Diagrams for an automated batch-dosing system

SKILLS

Software: SolidWorks (CSWA), MATLAB, AutoCAD, Ansys, MS Suite, LaTeX, Machine vision, Epicor, SAP

Industrial: Ignition, Studio 5000, C-More, Lean Concepts